

JAYCE(SEONGJIN) CHOI

MANUFACTURING SYSTEMS ENGINEER (MES / SMART FACTORY)

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PROFILE

Manufacturing Systems Engineer specializing in MES/MCS integration, smart factory deployment, and industrial automation for large-scale U.S. EV battery plants. Expert in leading equipment interface integration (CIM/MCS), SQL-driven material flow optimization (RTD/MHS), and deployment integrity automation. Proven track record of reducing verification time by 90% and ensuring 100% source-to-production consistency. Experienced in supporting 24/7 high-volume manufacturing across multi-site operations (OH, MI, GA).

PROFESSIONAL EXPERIENCE

Manufacturing Systems Engineer(MES), LG CNS America Jan 2023 to Present
Warren, OH / Holt, MI / Ellabell, GA

Played a key role in large-scale smart factory deployment for multi-site U.S. EV battery plants operating 24/7 high-throughput production environments.

- Led MES/MCS/CIM integration with automated material handling systems (conveyors, stacker cranes, AGVs/AMRs), enabling real-time material tracking across EV battery production lines.
- Architected and deployed a Python-based MES Deployment Diagnostic Framework using GitLab REST API and Regex validation logic, reducing pre-deployment verification time by 90%.
- Implemented automated 3-way reconciliation (GitLab source vs. Server API vs. Deployment list), ensuring 100% system integrity across BizActor and service IDs.
- Optimized RTD/MHS SQL logic to improve material routing accuracy, reduce dispatch latency, and enhance production stability.
- Conducted cross-layer system validation across MHS–WMS–ERP integrations to ensure end-to-end material traceability.
- Performed deep-dive root cause analysis of recurring MES/MCS production incidents within centralized IT shared service environments.
- Supported commissioning and on-site go-live stabilization for multiple U.S. battery plants operating in 24/7 high-throughput production environments.
- Collaborated with cross-functional teams including controls engineers, IT infrastructure, and production stakeholders to align automation logic with operational requirements.

EDUCATION

Hongik University Mar 2014 to Feb 2018
Bachelor of Computer Science, Seoul, South Korea

AREAS OF EXPERTISE

Manufacturing Systems Architecture (MES/MCS/CIM, RTD/MHS)	Oracle SQL Performance Engineering	Industrial Automation Protocols (PLC, OPC-UA, HSMS-GEM)	Event-Driven Distributed Systems (Solace, Pub/Sub Messaging)
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TECHNICAL PROJECT

MES Deployment & Source Integrity Diagnostic Tool

- Developed an automation solution using **Python** and **GitLab REST API** to validate consistency between source code changes and production server instances.
- Implemented a **3-way reconciliation logic** (GitLab vs. API vs. Deployment List) using **Regex** to ensure 100% integrity of BizActor and Service IDs.
- Achieved a **90% reduction** in manual verification time and prevented potential service omissions across multi-site battery plants.

Python Logistics Visualization Tool

- Developed a Python-based visualization tool to analyze logistics flow, routing patterns, equipment anomalies, and process performance using real MES/MCS/RTD data. Converted raw SQL outputs into clear graphical insights, enabling faster troubleshooting and data-driven decision-making for smart factory operations.